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Providing runtime power profile tuning based on eye state in processor-based extended reality (XR) devices

Abstract

Providing runtime power profile tuning based on eye state in processor-based Extended Reality (XR) devices is disclosed herein. In this regard, in some exemplary aspects, a processor of a processor-based XR device is configured to apply a default power profile, and detect, using an eye blink detection circuit, an eye close event of a user of the processor-based XR device. In response to detecting the eye close event, the processor selects a test power profile from a plurality of power profiles and applies the test power profile. The processor then measures an operating characteristic of the processor-based XR device and compares the operating characteristic with a corresponding reference range. Based on the comparing, the processor updates one or more power profiles of the plurality of power profiles.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10698479	12/2019	Fan	N/A	A61B 3/113
10921883	12/2020	Onyekwelu	N/A	G06F 3/013
2012/0320334	12/2011	Ho	351/159.23	G02C 7/044
2015/0039922	12/2014	Chalhoub	713/323	G06F 1/3296
2017/0039045	12/2016	Abrahami	N/A	A61B 5/486
2017/0178408	12/2016	Bavor, Jr. et al.	N/A	N/A
2017/0188823	12/2016	Ganesan	N/A	A61B 3/113
2017/0255259	12/2016	Mor	N/A	G06F 1/3265
2018/0329465	12/2017	Tavakoli	N/A	G06F 1/3234
2019/0163250	12/2018	Lee	N/A	G06F 11/3062
2020/0019238	12/2019	Agaoglu et al.	N/A	N/A
2020/0060810	12/2019	Toner	N/A	G02C 7/027
2020/0073122	12/2019	Rothkopf	N/A	G02B 27/017
2020/0073465	12/2019	Nikhara	N/A	G06F 1/324
2022/0000362	12/2021	Scheller	N/A	A61B 3/005
2022/0083121	12/2021	Khosrowpour et al.	N/A	N/A
2024/0139511	12/2023	Rangaraju	N/A	A61B 5/6803

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Patent Application No.

PCT/US2023/071423, mailed Nov. 7, 2023, 17 pages. cited by applicant

Benedetto, S. et al., "Driver workload and eye blink duration," Transportation Research Part F: Traffic Psychology and Behaviour, vol. 14, No. 3, May 2011, pp. 199-208. cited by applicant

Briggs, "How Long Is A "Blink of an Eye" Astronomically?" Jul. 2, 2017, available on the Internet: [URL: <https://www.wmbriggs.com/post/1750/>], 7 pages. cited by applicant

Wascher, E. et al., "Eye-blinks in choice response tasks uncover hidden aspects of information processing," EXCLI Journal, vol. 14, Nov. 2015, 12 pages. cited by applicant

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Background/Summary

BACKGROUND

I. Field of the Disclosure

(1) The technology of the disclosure relates generally to power management in processor-based Extended Reality (XR) devices.

II. Background

(2) Extended reality (XR), as used herein, is an umbrella term that refers to computer-generated environments and interactions that range from an immersive virtual environments of a virtual reality (VR) application to a combination of real-world elements and virtual elements provided by augmented reality (AR) and mixed reality (MR) applications. An XR environment is generated by a processor-based XR device, which may comprise any computing device that provides a processor, a display device (such as a screen, a head-mounted display (HMD), eyeglasses, and/or the like, as non-limiting examples), an input device, and appropriate sensors for detecting position and movement of the computing device, the input device, and/or the HMD (e.g., accelerometers, compasses, positioning systems, and/or the like, as non-limiting examples). Such processor-based XR devices can provide an immersive experience in which a user perceives an entirely virtual, computer-generated environment (in the case of VR), or an environment in which computer-generated digital information or virtual objects are overlaid onto a non-virtual real-world environment (in the case of AR/MR).

(3) Because the processing, sensing, and display functionalities provided by processor-based XR devices tend to be power-intensive, the optimization of power consumption is a challenge faced by most processor-based XR devices. One approach to managing power consumption involves the use of predefined power profiles that specify various system settings, options, and/or attributes that can be modified to optimize power usage in different use cases. Each use case may be associated with one or more specific power profiles, and may correspond to a particular mode of operation of the processor-based XR device (e.g., a full performance mode, a battery-saving mode, and the like), a particular class of application being executed by the processor-based XR device (e.g., a game application or other performance-intensive application), and/or a particular hardware configuration of the processor-based XR device.

(4) However, the use of power profiles may not be sufficient to adequately optimize power consumption by a processor-based XR device. The actual real-time power consumption for a given use case may not be constant, but rather may vary depending on variables such as workload, central processing unit (CPU) usage, cache usage, and/or other resource usage patterns. Power consumption may be further impacted by variations in design and development of third-party applications over different releases, as well as by differences between operating system versions. As a result, the usage of a particular power profile for a given use case over multiple occasions may result in different levels of power consumption, as well as different impacts on device performance and thermal profile, depending on the interactions of these variables. Moreover, testing different power profiles under runtime circumstances may risk negatively impacting the user experience by degrading device performance and/or thermal profile.

(5) Accordingly, a mechanism for more effectively tuning power profiles of processor-based XR devices for different use cases while minimizing impact on user experience is desirable.

SUMMARY OF THE DISCLOSURE

(6) Aspects disclosed in the detailed description include providing runtime power profile tuning based on eye state in processor-based Extended Reality (XR) devices. Related apparatus, methods, and computer-readable media are also disclosed. In this regard, in some exemplary aspects disclosed herein, a processor-based XR device (e.g., using a power profile tuning circuit) is configured to perform power profile testing and tuning during an “eye blink window” (i.e., an interval between detection of the closing of a user's eyes and detection of the subsequent opening of the user's eyes), during which the user will not perceive any performance degradation that may

result from power profile tuning. The processor-based XR device begins by applying a default power profile from a plurality of power profiles, based on, e.g., a current use case for the processor-based XR device or an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM)-provided whitelist of power profiles. The processor-based XR device then detects, using an eye blink detection circuit, an eye close event of the user. Upon detecting the eye close event, the processor-based XR device selects a test power profile from the plurality of power profiles and applies the test power profile. The processor-based XR device then measures an operating characteristic (e.g., a power characteristic, a performance characteristic, and/or a temperature characteristic, as non-limiting examples) of the processor-based XR device and compares the operating characteristic with a corresponding reference range (e.g., a reference power range, a reference performance range, and/or a reference temperature range). Based on the comparing, the processor-based XR device updates one or more power profiles of the plurality of power profiles (e.g., to increase or decrease the frequency with which the test power profile will be selected in the future, to indicate that the test power profile should be evaluated further, and/or to select a new default power profile, as non-limiting examples). In this manner, the effects of different power profiles on power consumption, device performance, and thermal profile can be dynamically evaluated and optimized at runtime in a manner that is transparent to the user, resulting in improved battery life and increased device usage time while minimizing impacts on performance and thermal profile.

(7) In some aspects, the processor-based XR device subsequently detects an eye open event of the user (indicating the end of the eye blink window), and, in response, reverts to the default power profile. Some aspects may provide that the processor-based XR device may measure a performance characteristic (e.g., motion-to-photon (M2P) latency, motion-to-render-to-photon (M2R2P) latency, frames per second (FPS), jank, six-degrees-of-freedom (6DOF)/perception drift, and/or network throughput, as non-limiting examples), and compare the performance characteristic to a reference performance range. If the performance characteristic is within the reference performance range, the processor-based XR device may then measure a power characteristic (e.g., average milliwatts (mW) consumed, as a non-limiting example), and compare the power characteristic to a reference power range. If the power characteristic is within the reference power range, the processor-based device updates the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window. According to some aspects, if the processor-based XR device determines that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile, the processor-based XR device may designate the test power profile as the default power profile.

(8) Some aspects may provide that the processor-based XR device may measure a temperature characteristic (e.g., junction temperature ($T_{sub.j}$) and/or skin temperature ($T_{sub.skin}$), as non-limiting examples) of the processor-based XR device and compare the temperature characteristic to a reference temperature range. If the temperature characteristic is within the reference temperature range, the processor-based device updates the test temperature profile to provide feedback for selecting the test power profile more frequently in the future. However, if the temperature characteristic is not within the reference temperature range, the processor-based XR device updates the test power profile to provide feedback for selecting the test power profile less frequently in the future, and also selects and applies a new power profile without waiting for an eye open event.

(9) In another aspect, a processor-based XR device is provided. The processor-based XR device comprises a processor, and an eye blink detection circuit communicatively coupled to the processor. The processor is configured to apply a default power profile. The processor is further configured to detect, using the eye blink detection circuit, an eye close event of a user of the processor-based XR device. The processor is also configured to, responsive to detecting the eye close event, select a test power profile from a plurality of power profiles, and apply the test power

profile. The processor is additionally configured to measure an operating characteristic of the processor-based XR device. The processor is further configured to compare the operating characteristic with a corresponding reference range. The processor is also configured to update one or more power profiles of the plurality of power profiles based on the comparing.

(10) In another aspect, a processor-based XR device is provided. The processor-based XR device comprises means for applying a default power profile. The processor-based XR device further comprises means for detecting an eye close event of a user of the processor-based XR device. The processor-based XR device also comprises means for selecting a test power profile from a plurality of power profiles, responsive to detecting the eye close event. The processor-based XR device additionally comprises means for applying the test power profile. The processor-based XR device further comprises means for measuring an operating characteristic of the processor-based XR device. The processor-based XR device also comprises means for comparing the operating characteristic with a corresponding reference range. The processor-based XR device additionally comprises means for updating one or more power profiles of the plurality of power profiles based on the comparing.

(11) In another aspect, a method for providing runtime power profile tuning based on eye state in processor-based XR devices is provided. The method comprises applying, by a power profile tuning circuit of a processor-based XR device, a default power profile. The method further comprises detecting, using an eye blink detection circuit of the processor-based XR device, an eye close event of a user of the processor-based XR device. The method also comprises, responsive to detecting the eye close event, selecting a test power profile from a plurality of power profiles, and applying the test power profile. The method additionally comprises measuring an operating characteristic of the processor-based XR device. The method further comprises comparing the operating characteristic with a corresponding reference range. The method also comprises updating one or more power profiles of the plurality of power profiles based on the comparing.

(12) In another aspect, a non-transitory computer-readable medium is provided. The non-transitory computer-readable medium stores thereon computer-executable instructions that, when executed by a processor of a processor-based XR device, cause the processor to apply a default power profile. The computer-executable instructions further cause the processor to detect an eye close event of a user of the processor-based XR device. The computer-executable instructions also cause the processor to, responsive to detecting the eye close event, select a test power profile from a plurality of power profiles, and apply the test power profile. The computer-executable instructions additionally cause the processor to measure an operating characteristic of the processor-based XR device. The computer-executable instructions further cause the processor to compare the operating characteristic with a corresponding reference range. The computer-executable instructions also cause the processor to update one or more power profiles of the plurality of power profiles based on the comparing.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a block diagram of an exemplary processor-based Extended Reality (XR) device configured to provide runtime power profile tuning based on eye state, according to some aspects;
- (2) FIG. 2 is a block diagram illustrating an exemplary eye blink timeline showing exemplary operations performed within an eye blink window;
- (3) FIGS. 3A and 3B are flowcharts illustrating exemplary operations for providing runtime power profile tuning based on eye state, according to some aspects;
- (4) FIGS. 4A and 4B are flowcharts illustrating further exemplary operations for providing runtime power profile tuning further based on performance and power characteristics of the processor-based

XR device when using a test power profile, according to some aspects;

(5) FIG. 5 is a flowchart illustrating exemplary operations for providing runtime power profile tuning further based on temperature characteristics of the processor-based XR device when using a test power profile, according to some aspects; and

(6) FIG. 6 is a block diagram of an exemplary processor-based device that can include the processor-based XR device of FIG. 1.

DETAILED DESCRIPTION

(7) With reference now to the drawing figures, several exemplary aspects of the present disclosure are described. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects.

(8) Aspects disclosed in the detailed description include providing runtime power profile tuning based on eye state in processor-based Extended Reality (XR) devices. Related apparatus, methods, and computer-readable media are also disclosed. In this regard, in some exemplary aspects disclosed herein, a processor-based XR device (e.g., using a power profile tuning circuit) is configured to perform power profile testing and tuning during an “eye blink window” (i.e., an interval between detection of the closing of a user's eyes and detection of the subsequent opening of the user's eyes), during which the user will not perceive any performance degradation that may result from power profile tuning. The processor-based XR device begins by applying a default power profile from a plurality of power profiles, based on, e.g., a current use case for the processor-based XR device or an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM)-provided whitelist of power profiles. The processor-based XR device then detects, using an eye blink detection circuit, an eye close event of the user. Upon detecting the eye close event, the processor-based XR device selects a test power profile from the plurality of power profiles and applies the test power profile. The processor-based XR device then measures an operating characteristic (e.g., a power characteristic, a performance characteristic, and/or a temperature characteristic, as non-limiting examples) of the processor-based XR device and compares the operating characteristic with a corresponding reference range (e.g., a reference power range, a reference performance range, and/or a reference temperature range). Based on the comparing, the processor-based XR device updates one or more power profiles of the plurality of power profiles (e.g., to increase or decrease the frequency with which the test power profile will be selected in the future, to indicate that the test power profile should be evaluated further, and/or to select a new default power profile, as non-limiting examples). In this manner, the effects of different power profiles on power consumption, device performance, and thermal profile can be dynamically evaluated and optimized at runtime in a manner that is transparent to the user, resulting in improved battery life and increased device usage time while minimizing impacts on performance and thermal profile.

(9) In some aspects, the processor-based XR device subsequently detects an eye open event of the user (indicating the end of the eye blink window), and, in response, reverts to the default power profile. Some aspects may provide that the processor-based XR device may measure a performance characteristic (e.g., motion-to-photon (M2P) latency, motion-to-render-to-photon (M2R2P) latency, frames per second (FPS), jank, six-degrees-of-freedom (6DOF)/perception drift, and/or network throughput, as non-limiting examples), and compare the performance characteristic to a reference performance range. If the performance characteristic is within the reference performance range, the processor-based XR device may then measure a power characteristic (e.g., average milliwatts (mW) consumed, as a non-limiting example), and compare the power characteristic to a reference power range. If the power characteristic is within the reference power range, the processor-based device updates the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window. According to some aspects, if the processor-based XR device determines that the test power profile has been evaluated over multiple eye blink

windows and indicates one of a performance improvement and a power improvement over the default power profile, the processor-based XR device may designate the test power profile as the default power profile.

(10) Some aspects may provide that the processor-based XR device may measure a temperature characteristic (e.g., junction temperature ($T_{sub.j}$) and/or skin temperature ($T_{sub.skin}$), as non-limiting examples) of the processor-based XR device and compare the temperature characteristic to a reference temperature range. If the temperature characteristic is within the reference temperature range, the processor-based device updates the test temperature profile to provide feedback for selecting the test power profile more frequently in the future. However, if the temperature characteristic is not within the reference temperature range, the processor-based XR device updates the test power profile to provide feedback for selecting the test power profile less frequently in the future, and also selects and applies a new power profile without waiting for an eye open event.

(11) In this regard, FIG. 1 is a block diagram of an exemplary processor-based XR device **100** that is configured to generate and display an XR environment (e.g., a virtual reality (VR) environment, an augmented reality (AR) environment, and/or a mixed reality (MR) environment) to a user **102**, and to update the displayed XR environment in response to input from the user **102** and/or changes in the XR environment. The processor-based XR device **100** comprises a processor **104** (also referred to as a “processor core” or a “central processing unit (CPU) core”), which may be an in-order or an out-of-order processor (OoP) and/or may be one of a plurality of processors **104** provided by the processor-based XR device **100**. In the example of FIG. 1, the processor **104** is communicatively coupled to a graphics processing unit (GPU) **106**, which is a dedicated processing unit configured to accelerate the rendering of graphics and video data for display by a display device **108**. The display device **108** may comprise, e.g., a head-mounted display (HMD), a monitor, eyeglasses, and/or the like. The processor **104** is also communicatively coupled to position/motion sensors **110** that are configured to provide the processor **104** with data regarding the position and motion of the user **102** and/or the processor-based XR device **100**. The position/motion sensors **110** may comprise, e.g., accelerometers, compasses, positioning systems, and/or the like, as non-limiting examples. The processor **104** is additionally communicatively coupled to a memory device **112** that provides storage for instructions and data used by the processor **104** when executing an XR application.

(12) The processor-based XR device **100** of FIG. 1 may encompass any one of known digital logic elements, semiconductor circuits, processing cores, and/or memory structures, among other elements, or combinations thereof. Aspects described herein are not restricted to any particular arrangement of elements, and the disclosed techniques may be easily extended to various structures and layouts on semiconductor dies or packages. It is to be understood that some aspects of the processor-based XR device **100** may include elements in addition to those illustrated in FIG. 1, and/or may include more or fewer of the elements illustrated in FIG. 1. For example, the processor-based XR device **100** may further include caches, controllers, communications buses, and/or persistent storage devices, which are omitted from FIG. 1 for the sake of clarity.

(13) The processor **104** of the processor-based XR device **100** provides an eye blink detection circuit **114** for monitoring the eye activity of the user **102**, including eye close events (i.e., the eyes of the user **102** transition from open to closed, such as when the user **102** begins an eye blink) and eye open events (i.e., the eyes of the user **102** transition from closed to open, as when the user **102** completes an eye blink). The eye blink detection circuit **114** may be based on convention eye tracking solutions such as those provided by Tobii Tech AB. To monitor the internal state of the processor-based XR device **100**, the processor **104** also includes a performance monitoring circuit **116**, an Embedded Power Monitor (EPM) **118**, and a temperature monitoring circuit **120**. The performance monitoring circuit **116** is configured to monitor and collect statistics on one or more performance characteristics that may affect the experience of the user **102** when using the processor-based XR device **100**, such as motion-to-photon (M2P) latency, motion-to-render-to-

photon (M2R2P) latency, frames per second (FPS), jank, six-degrees-of-freedom (6DOF)/perception drift, and/or network throughput, as non-limiting examples. The EPM **118** is configured to provide power measurement for all Power Management Integrated Circuit (PMIC) regulators (not shown) of the processor-based XR device **100** and to subsample each regulator during configurable aperture windows and provide sufficient accuracy for thermal control and system power budgeting by the processor-based XR device **100**. The temperature monitoring circuit **120** is configured to monitor and collect statistics for one or more of a junction temperature $T_{sub,j}$ and a skin temperature $T_{sub,skin}$ of the processor-based XR device **100**.

(14) To manage power consumption by the processor-based XR device **100**, the processor **104** provides a plurality of predefined power profiles **122(0)-122(P)** that are stored in a power profile data store **124**, and that are each associated with one or more use cases of a plurality of use cases **126(0)-126(U)** stored in a use case data store **128**. It is to be understood that, while the use cases **126(0)-126(U)** are shown in FIG. 1 as being stored separately from the power profiles **122(0)-122(P)**, some aspects may provide that the use cases **126(0)-126(U)** may be stored along with the power profiles **122(0)-122(P)** (e.g., as a use case indicator associated with each entry of the power profile data store **124**) rather than stored separately in the use case data store **128**. The use case data store **128** and the power profile data store **124** may comprise any appropriate data structure, data base, or lookup table (LUT).

(15) Each of the use cases **126(0)-126(U)** may correspond to one or more of a particular mode of operation of the processor-based XR device **100** (such as a full performance mode, a battery-saving mode, and the like, as non-limiting examples), a particular class of application being executed by the processor-based XR device **100** (such as a game application or other performance-intensive application, as non-limiting examples), and/or a particular hardware configuration of the processor-based XR device **100**. The power profiles **122(0)-122(P)** each specify one or more system settings, options, and/or attributes that can be implemented to optimize power usage in different ones of the use cases **126(0)-126(U)**. The power profiles **122(0)-122(P)** may also include additional information, such as an identifier of a corresponding use case of the use cases **126(0)-126(U)**, data indicating how frequently or under what conditions the power profiles **122(0)-122(P)** should be applied, and/or data indicating whether the power profiles **122(0)-122(P)** are being evaluated.

(16) Exemplary power profiles are shown below in Table 1:

(17) TABLE-US-00001 TABLE 1 Power Profile System Setting/Option/Attribute Modified Scale
Down System clock or specified subsystem is scaled down Skip Frames Skip processing of one or more frames of XR content Foveation Increase foveation region(s) scale from peripheral for Level certain frames Resolution Lower rendering resolution scaler Scaler System Memory Portion of printed circuit board (PCB)/bank/segment of Collapsible cache or memory is collapsible Collapse or Pause Collapse or pause certain hardware activity, such as Hardware camera Activity CPU tuning Modify scheduler or frequency of CPU Prefetcher Modify prefetcher setting, such as settings for Level 1 configuration (L1) cache, Level 2 (L2) cache, or Level 3 (L3) cache prefetcher DDR Idle Time Idle Power Down Count (IPDC) stats for memory Workload Move workload from a higher power processor or functional unit to lower power processor or functional unit

(18) However, as noted above, the use of power profiles such as the power profiles **122(0)-122(P)** of FIG. 1 may not be sufficient to adequately optimize power consumption by the processor-based XR device **100**, due to variations in workload, CPU usage, and cache usage at runtime, as well as the design and development of third-party applications over different releases and differences between operating system versions. Consequently, the usage of a particular power profile **122(0)-122(P)** for a given use case **126(0)-126(U)** over multiple occasions may result in different levels of power consumption, as well as different impacts on device performance and thermal profile, depending on the interactions of these variables.

(19) In this regard, the processor-based XR device **100** provides a power profile tuning circuit **130** that is configured to perform power profile testing and tuning based on an eye state of the user **102**.

In particular, the power profile tuning circuit **130** performs power profile tuning during an “eye blink window” (i.e., an interval between detection of the closing of a user's eyes and detection of the subsequent opening of the user's eyes). Because performance degradation that may result from the power profile tuning circuit **130** performing tuning operations during an eye blink window will not be perceived by the user **102**, power profile tuning can be performed at runtime, and thus can better account for the variations noted above.

(20) In exemplary operation, the power profile tuning circuit **130** in some aspects selects one of the power profiles **122(0)-122(P)** as a default power profile, based on, e.g., a use case of the plurality of use cases **126(0)-126(U)** and/or a whitelist provided by an OEM/ODM. In the example of FIG. **1**, assume for the sake of illustration that the power profile **122(0)** is selected as the default power profile. The power profile tuning circuit **130** may then apply the default power profile **122(0)** (i.e., by setting the system settings, options, and/or attributes to those indicated by the default power profile **122(0)**).

(21) Subsequently, the power profile tuning circuit **130** performs power profile tuning by using the eye blink detection circuit **114** to detect an eye close event of the user **102**. In response to the eye close event, the power profile tuning circuit **130** selects a test power profile, such as the power profile **122(1)**, from the plurality of power profiles **122(0)-122(P)**. The power profile tuning circuit **130** then applies the test power profile **122(1)** (i.e., by changing the system settings, options, and/or attributes from those indicated by the default power profile **122(0)** to those indicated by the test power profile **122(1)**). The power profile tuning circuit **130** then measures an operating characteristic of the processor-based XR device **100** while it operates under the test power profile **122(1)**. The operating characteristic may comprise a performance characteristic (captioned as “PERF CHAR” in FIG. **1**) **132** measured by the performance monitoring circuit **116**, a power characteristic (captioned as “POWER CHAR” in FIG. **1**) **134** measured by the EPM **118**, and/or a temperature characteristic (captioned as “TEMP CHAR” in FIG. **1**) **136** measured by the temperature monitoring circuit **120**.

(22) The power profile tuning circuit **130** next compares the operating characteristic with a corresponding reference range. The reference range specifies an acceptable range of values for the operating characteristic, and may comprise a reference performance range **138**, a reference power range **140**, and/or a reference temperature range **142** maintained by the power profile tuning circuit **130**. The power profile tuning circuit **130** then updates one or more power profiles of the plurality of power profiles **122(0)-122(P)** based on the comparing. As discussed in greater detail below, updating a power profile may comprise providing information regarding how frequently the test power profile **122(1)** is to be applied in the future, information regarding whether the test power profile **122(1)** is currently under evaluation, and/or providing other information for guiding future selection of the test power profile **122(1)**. In some aspects, the power profile tuning circuit **130** subsequently uses the eye blink detection circuit **114** to detect an eye open event of the user **102**, and, in response, reverts to the default power profile **122(0)**.

(23) According to some aspects, the specific operations for performing power profile tuning may include the power profile tuning circuit **130** measuring the performance characteristic **132** and comparing the performance characteristic **132** with the reference performance range **138**. If the performance characteristic **132** is not within the reference performance range **138**, the power profile tuning circuit **130** reverts to the default power profile **122(0)**. However, if the power profile tuning circuit **130** determines that the performance characteristic **132** is within the reference performance range **138**, the power profile tuning circuit **130** next measures the power characteristic **134** and compares the power characteristic **134** with the reference power range **140**. If the power characteristic **134** is not within the reference power range **140**, the power profile tuning circuit **130** reverts to the default power profile **122(0)**. If the power profile tuning circuit **130** determines that the power characteristic **134** is within the reference power range **140**, though, the power profile tuning circuit **130** updates the test power profile **122(1)** to indicate that evaluation of the test power

profile **122(1)** should continue during a subsequent eye blink window. Subsequently, after a number of eye blink windows have elapsed, the power profile tuning circuit **130** according to some aspects may determine whether the test power profile **122(1)** has been evaluated over multiple eye blink windows, and further whether the test power profile **122(1)** indicates a performance improvement and/or a power improvement over the default power profile **122(0)**. If so, the power profile tuning circuit **130** designates the test power profile **122(1)** as the default power profile **122(0)**.

(24) In some aspects, the specific operations for performing power profile tuning may include the power profile tuning circuit **130** measuring the temperature characteristic **136** (e.g., a junction temperature $T_{sub,j}$ and/or a skin temperature $T_{sub,skin}$ of the processor-based XR device **100**) and comparing the temperature characteristic **136** with the reference temperature range **142**. If the temperature characteristic **136** is within the reference temperature range **142**, the power profile tuning circuit **130** updates the test power profile **122(1)** to provide feedback for selecting the test power profile **122(1)** more frequently in the future. However, if the power profile tuning circuit **130** determines that the temperature characteristic **136** is not within the reference temperature range **142**, the power profile tuning circuit **130** may update the test power profile **122(1)** to provide feedback for selecting the test power profile **122(1)** less frequently in the future. The power profile tuning circuit **130** may also select a new power profile, such as the power profile **122(P)**, from the plurality of power profiles **122(0)-122(P)** and may apply the new power profile **122(P)** without waiting for an eye open event.

(25) Some aspects may provide that, prior to beginning power profile tuning, the power profile tuning circuit **130** first measures an initial operating characteristic (e.g., an initial performance characteristic such as the performance characteristic **132**, an initial power characteristic such as the power characteristic **134**, and/or an initial temperature characteristic such as the temperature characteristic **136**) of the processor-based XR device **100** while the processor-based XR device **100** is operating using the default power profile **122(0)**. The power profile tuning circuit **130** may then determine whether the initial operating characteristic of the processor-based XR device **100** is within a corresponding reference range (e.g., the reference performance range **138**, the reference power range **140**, and/or the reference temperature range **142**). For example, the power profile tuning circuit **130** may determine whether an average power usage in milliwatts is within a reference milliwatt range, whether a junction temperature $T_{sub,j}$ is within a reference temperature range, and whether a Key Performance Indicator (KPI) is within a reference performance range. If not, the power profile tuning circuit **130** updates the default power profile **122(0)** to provide feedback for selecting the default power profile **122(0)** less frequently in the future. The power profile tuning circuit **130** may select a new power profile, such as the power profile **122(P)**, from the plurality of power profiles **122(0)-122(P)**. The power profile tuning circuit **130** then applies the new power profile **122(P)**.

(26) FIG. 2 is a block diagram illustrating an exemplary eye blink timeline **200** along with exemplary operations that may be performed within an eye blink window by the processor-based XR device **100** of FIG. 1. On the eye blink timeline **200**, multiple eye blink windows **202(0)-202(2)** are shown, with each of the eye blink windows **202(0)-202(2)** beginning with a corresponding eye close event **204(0)-204(2)** and ending with a corresponding eye open event **206(0)-206(2)**. Eye blink windows such as the eye blink windows **202(0)-202(2)** may occur from six (6) to 30 times per minute on average, with each of the eye blink windows **202(0)-202(2)** lasting approximately 300 to 400 milliseconds. As seen in FIG. 2, during the eye blink window **202(1)**, the power profile tuning circuit **130**, upon detecting the eye close event **204(1)**, may select and apply the test power profile **122(1)** as described above and represented by block **208**. The power profile tuning circuit **130** then measures an operating characteristic of the processor-based XR device **100** while it operates under the test power profile **122(1)**, as represented by block **210** (captioned as “MEASURE CHAR” in FIG. 2), and compares the operating characteristic with a corresponding

reference range, as represented by block **212** (captioned as “COMPARE CHAR” in FIG. 2). The power profile tuning circuit **130** then updates one or more power profiles of the plurality of power profiles **122(0)-122(P)** based on the comparing, as represented by block **214** (captioned as “UPDATE PROFILE” in FIG. 2). Upon detecting the eye open event **206(1)**, the power profile tuning circuit **130** reverts to the default power profile **122(0)**, as indicated by block **216** (captioned as “REVERT PROFILE” in FIG. 2).

(27) To illustrating exemplary operations for providing runtime power profile tuning based on eye state in the processor-based XR device **100** of FIG. 1, FIGS. 3A and 3B provide flowcharts illustrating exemplary operations **300**. For the sake of clarity, elements of FIGS. 1 and 2 are referenced in describing FIGS. 3A and 3B. It is to be understood that some of the operations **300** shown in FIGS. 3A and 3B may be performed in an order other than that illustrated herein, and/or may be omitted. The operations **300** begin in FIG. 3A with the processor **104** in some aspects selecting a default power profile (e.g., the power profile **122(0)** of FIG. 1) based on one or more of a use case (e.g., the use case **126(0)** of FIG. 1) of the processor-based XR device **100** and a whitelist provided by an OEM/ODM (block **302**). The processor **104** (e.g., using the power profile tuning circuit **130** of FIG. 1) applies the default power profile **122(0)** (block **304**).

(28) According to some aspects, the processor **104** may measure an initial operating characteristic (e.g., an initial performance characteristic such as the performance characteristic **132** of FIG. 1, an initial power characteristic such as the power characteristic **134** of FIG. 1, and/or an initial temperature characteristic such as the temperature characteristic **136** of FIG. 1) of the processor-based XR device **100** (block **306**). The processor **104** then determines whether the initial operating characteristic of the processor-based XR device **100** is within a corresponding reference range (e.g., the reference performance range **138**, the reference power range **140**, and/or the reference temperature range **142** of FIG. 1) (block **308**). If so, processing continues at block **310** of FIG. 3B. If the initial operating characteristic is determined at decision block **308** to not be within the corresponding reference range, the processor **104** updates the default power profile **122(0)** to provide feedback for selecting the default power profile **122(0)** in the future (e.g., so the default power profile **122(0)** will be selected less frequently) (block **312**). The processor **104** next selects a new power profile (e.g., the power profile **122(P)** of FIG. 1) from the plurality of power profiles **122(0)-122(P)** (block **314**). The processor **104** then applies the new power profile **122(P)** (block **316**). The operations **300** then continue at block **316** of FIG. 3B.

(29) Turning now to FIG. 3B, the operations **300** continue with the processor **104** detecting, using an eye blink detection circuit (e.g., the eye blink detection circuit **114** of FIG. 1) of the processor **104**, an eye close event (e.g., the eye close event **204(1)** of FIG. 2) of a user (e.g., the user **102** of FIG. 1) of the processor-based XR device **100** (block **310**). The processor **104** then performs a series of operations responsive to detecting the eye close event **204(1)** (block **318**). The processor **104** selects a test power profile (e.g., the power profile **122(1)** of FIG. 1) from a plurality of power profiles (e.g., the profiles **122(0)-122(P)** of FIG. 1) (block **320**). The processor **104** then applies the test power profile **122(1)** (block **322**). The processor **104** measures an operating characteristic (e.g., the performance characteristic **132**, the power characteristic **134**, and/or the temperature characteristic **136** of FIG. 1) of the processor-based XR device **100** (block **324**). The processor **104** compares the operating characteristic with a corresponding reference range (e.g., the reference performance range **138**, the reference power range **140**, and/or the reference temperature range **142** of FIG. 1) (block **326**). The processor **104** then updates one or more power profiles of the plurality of power profiles **122(0)-122(P)** based on the comparing (block **328**). In some aspects, the processor **104** may subsequently detect, using the eye blink detection circuit **114**, an eye open event (e.g., the eye open event **206(1)** of FIG. 2) of the user **102** of the processor-based XR device **100** (block **330**). In response to detecting the eye open event **206(1)**, the processor **104** may revert to the default power profile **122(0)** (block **332**).

(30) FIGS. 4A and 4B are flowcharts that illustrate exemplary operations **400** for providing runtime

power profile tuning further based on performance and power characteristics of the processor-based XR device **100** when using a test power profile such as the power profile **122(1)** of FIG. **1**.

Elements of FIGS. **1** and **2** are referenced in describing FIGS. **4A** and **4B** for the sake of clarity. In FIG. **4A**, the operations **400** begin with the processor **104** (e.g., using the power profile tuning circuit **130** of FIG. **1**) measuring a performance characteristic (e.g., the performance characteristic **132** of FIG. **1**) of the processor-based XR device **100** (block **402**). In this regard, the operations of block **402** of FIG. **4A** correspond to the operations of block **324** of FIG. **3B** for measuring an operating characteristic of the processor-based XR device **100** in some aspects. The processor **104** then compares the performance characteristic **132** with a reference performance range, such as the reference performance range **138** of FIG. **1** (block **404**). The operations of block **404** of FIG. **4A** thus correspond to the operations of block **326** of FIG. **3B** for comparing the operating characteristic with a corresponding reference range, according to some aspects.

(31) The processor **104** next determines whether the performance characteristic **132** is within the reference performance range **138** (block **406**). If not, the processor **104** reverts to the default power profile **122(0)** (block **408**). However, if the processor **104** determines at decision block **406** that the performance characteristic **132** is within the reference performance range **138**, the processor **104** measures a power characteristic (e.g., the power characteristic **134** of FIG. **1**) of the processor-based XR device **100** (block **410**). The processor **104** then compares the power characteristic **134** with a reference power range (e.g., the reference power range **140** of FIG. **1**) (block **412**). The operations **400** then continue at block **414** of FIG. **4B**.

(32) Referring now to FIG. **4B**, the operations **400** continue with the processor **104** determining whether the power characteristic **134** is within the reference power range **140** (block **414**). If not, the processor **104** reverts to the default power profile **122(0)** (block **416**). However, if the processor **104** determines at decision block **414** that the power characteristic **134** is within the reference power range **140**, the processor **104** updates the test power profile **122(1)** to indicate that evaluation of the test power profile **122(1)** should continue during a subsequent eye blink window **202(2)** (block **418**). In this regard, the operations of blocks **406-418** of FIGS. **4A** and **4B** correspond to the operations of block **328** of FIG. **3B** for updating one or more power profiles of the plurality of power profiles **122(0)-122(P)** based on the comparing.

(33) In some aspects, the processor **104** may also determine whether the test power profile **122(1)** has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile **122(0)** (block **420**). If so, the processor **104** designates the test power profile **122(1)** as the default power profile **122(0)** (block **422**). If it is determined at decision block **420** that the test power profile **122(1)** has not been evaluated over multiple eye blink windows or does not indicate a performance improvement or a power improvement over the default power profile **122(0)**, the processor **104** continues evaluation of test power profile **122(1)** during a subsequent eye blink window **202(2)** (block **424**).

(34) To illustrate exemplary operations for providing runtime power profile tuning further based on temperature characteristics of the processor-based XR device **100** of FIG. **1** when using a test power profile such as the power profile **122(1)** of FIG. **1**, FIG. **5** provides a flowchart showing exemplary operations **500**. For the sake of clarity, elements of FIGS. **1** and **2** are referenced in describing FIG. **5**. The operations **500** begin in FIG. **5A** with the processor **104** of FIG. **1** (e.g., using the power profile tuning circuit **130** of FIG. **1**) measuring a temperature characteristic (e.g., the temperature characteristic **136** of FIG. **1**) of the processor-based XR device **100** (block **502**). In this regard, the operations of block **502** of FIG. **5** correspond to the operations of block **324** of FIG. **3B** for measuring an operating characteristic of the processor-based XR device **100** in some aspects. The processor **104** next compares the temperature characteristic **136** with a reference temperature range (e.g., the reference temperature range **142** of FIG. **1**) (block **504**). The operations of block **504** of FIG. **5** thus correspond to the operations of block **326** of FIG. **3B** for comparing the operating characteristic with a corresponding reference range, according to some aspects.

(35) The processor **104** then determines whether the temperature characteristic **136** is within the reference temperature range **142** (block **506**). If so, the processor **104** updates the test power profile **122(1)** to provide feedback for selecting the test power profile **122(1)** more frequently in the future (block **508**). However, if the processor **104** determines at decision block **506** that the temperature characteristic **136** is not within the reference temperature range **142**, the processor **104** updates the test power profile **122(1)** to provide feedback for selecting the test power profile **122(1)** less frequently in the future (block **510**). The processor **104** selects a new power profile (e.g., the power profile **122(P)** of FIG. **1**) from the plurality of power profiles **122(0)-122(P)** (block **512**). The processor **104** then applies the new power profile **122(P)** without waiting for an eye open event (e.g., the eye open event **206(1)** of FIG. **2**) (block **514**). In this regard, the operations of blocks **506-514** of FIG. **5** correspond to the operations of block **328** of FIG. **3B** for updating one or more power profiles of the plurality of power profiles **122(0)-122(P)** based on the comparing.

(36) Providing runtime power profile tuning based on eye state according to aspects disclosed herein may be provided in or integrated into any processor-based device. Examples, without limitation, include a set top box, an entertainment unit, a navigation device, a communications device, a fixed location data unit, a mobile location data unit, a global positioning system (GPS) device, a mobile phone, a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a tablet, a phablet, a server, a computer, a portable computer, a mobile computing device, laptop computer, a wearable computing device (e.g., a smart watch, a health or fitness tracker, eyewear, etc.), a desktop computer, a personal digital assistant (PDA), a monitor, a computer monitor, a television, a tuner, a radio, a satellite radio, a music player, a digital music player, a portable music player, a digital video player, a video player, a digital video disc (DVD) player, a portable digital video player, an automobile, a vehicle component, an avionics system, a drone, and a multicopter.

(37) In this regard, FIG. **6** illustrates an example of a processor-based device **600** that may comprise the processor-based XR device **100** illustrated in FIG. **1**. In this example, the processor-based device **600** includes a processor **602** that includes one or more central processing units (captioned as “CPUs” in FIG. **6**) **604**, which may also be referred to as CPU cores or processor cores. The processor **602** may have cache memory **606** coupled to the processor **602** for rapid access to temporarily stored data. The processor **602** is coupled to a system bus **608** and can intercouple master and slave devices included in the processor-based device **600**. As is well known, the processor **602** communicates with these other devices by exchanging address, control, and data information over the system bus **608**. For example, the processor **602** can communicate bus transaction requests to a memory controller **610**, as an example of a slave device. Although not illustrated in FIG. **6**, multiple system buses **608** could be provided, wherein each system bus **608** constitutes a different fabric.

(38) Other master and slave devices can be connected to the system bus **608**. As illustrated in FIG. **6**, these devices can include a memory system **612** that includes the memory controller **610** and a memory array(s) **614**, one or more input devices **616**, one or more output devices **618**, one or more network interface devices **620**, and one or more display controllers **622**, as examples. The input device(s) **616** can include any type of input device, including but not limited to input keys, switches, voice processors, etc. The output device(s) **618** can include any type of output device, including, but not limited to, audio, video, other visual indicators, etc. The network interface device(s) **620** can be any device configured to allow exchange of data to and from a network **624**. The network **624** can be any type of network, including, but not limited to, a wired or wireless network, a private or public network, a local area network (LAN), a wireless local area network (WLAN), a wide area network (WAN), a BLUETOOTH™ network, and the Internet. The network interface device(s) **620** can be configured to support any type of communications protocol desired.

(39) The processor **602** may also be configured to access the display controller(s) **622** over the system bus **608** to control information sent to one or more displays **626**. The display controller(s) **622** sends information to the display(s) **626** to be displayed via one or more video processors **628**,

which process the information to be displayed into a format suitable for the display(s) **626**. The display(s) **626** can include any type of display, including but not limited to a cathode ray tube (CRT), a liquid crystal display (LCD), a plasma display, etc.

(40) Those of skill in the art will further appreciate that the various illustrative logical blocks, modules, circuits, and algorithms described in connection with the aspects disclosed herein may be implemented as electronic hardware, instructions stored in memory or in another computer readable medium and executed by a processor or other processing device, or combinations of both. Memory disclosed herein may be any type and size of memory and may be configured to store any type of information desired. To clearly illustrate this interchangeability, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. How such functionality is implemented depends upon the particular application, design choices, and/or design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

(41) The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(42) The aspects disclosed herein may be embodied in hardware and in instructions that are stored in hardware, and may reside, for example, in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, a hard disk, a removable disk, a CD-ROM, or any other form of computer readable medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a remote station. In the alternative, the processor and the storage medium may reside as discrete components in a remote station, base station, or server.

(43) It is also noted that the operational steps described in any of the exemplary aspects herein are described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(44) The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus,

the disclosure is not intended to be limited to the examples and designs described herein, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

(45) Implementation examples are described in the following numbered clauses:

(46) 1. A processor-based Extended Reality (XR) device, comprising: a processor; and an eye blink detection circuit communicatively coupled to the processor; the processor configured to: apply a default power profile; detect, using the eye blink detection circuit, an eye close event of a user of the processor-based XR device; and responsive to detecting the eye close event: select a test power profile from a plurality of power profiles; apply the test power profile; measure an operating characteristic of the processor-based XR device; compare the operating characteristic with a corresponding reference range; and update one or more power profiles of the plurality of power profiles based on the comparing.

(47) 2. The processor-based XR device of clause 1, wherein the processor is further configured to select the default power profile based on one or more of a use case of the processor-based XR device and a whitelist provided by an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM).

(48) 3. The processor-based XR device of any one of clauses 1-2, wherein the processor is further configured to: detect, using the eye blink detection circuit, an eye open event of the user of the processor-based XR device; and responsive to detecting the eye open event, revert to the default power profile.

(49) 4. The processor-based XR device of clause 3, wherein: the processor is configured to measure the operating characteristic of the processor-based XR device by being configured to measure a performance characteristic of the processor-based XR device; the processor is configured to compare the operating characteristic with the corresponding reference range by being configured to compare the performance characteristic with a reference performance range; and the processor is configured to update one or more power profiles of the plurality of power profiles based on the comparing by being configured to: determine whether the performance characteristic is within the reference performance range; and responsive to determining that the performance characteristic is within the reference performance range: measure a power characteristic of the processor-based XR device; compare the power characteristic with a reference power range; determine whether the power characteristic is within the reference power range; and responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event, update the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window.

(50) 5. The processor-based XR device of any one of clauses 3-4, wherein the processor is further configured to: determine whether the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile, designate the test power profile as the default power profile.

(51) 6. The processor-based XR device of any one of clauses 3-4, wherein the processor is further configured to, responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event, revert to the default power profile.

(52) 7. The processor-based XR device of any one of clauses 3-4, wherein the processor is further configured to, responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event, revert to the default power profile.

(53) 8. The processor-based XR device of any one of clauses 1-7, wherein: the processor is configured to measure the operating characteristic of the processor-based XR device by being

configured to measure a temperature characteristic of the processor-based XR device; the processor is configured to compare the operating characteristic with the corresponding reference range by being configured to compare the temperature characteristic with a reference temperature range; and the processor is configured to update one or more power profiles of the plurality of power profiles based on the comparing by being configured to: determine whether the temperature characteristic is within the reference temperature range; and responsive to determining that the temperature characteristic is not within the reference temperature range: update the test power profile to provide feedback for selecting the test power profile less frequently in the future; select a new power profile from the plurality of power profiles; and apply the new power profile without waiting for an eye open event.

(54) 9. The processor-based XR device of any one of clauses 1-8, wherein the processor is further configured to, prior to applying the default power profile: measure an initial operating characteristic of the processor-based XR device; and determine whether the initial operating characteristic of the processor-based XR device is within the corresponding reference range; wherein the processor is configured to detect the eye close event of the user of the processor-based XR device responsive to determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range.

(55) 10. The processor-based XR device of clause 9, wherein the processor is configured to determine whether the initial operating characteristic of the processor-based XR device is within the corresponding reference range by being configured to: determine whether a performance characteristic is within a reference performance range; determine whether a power characteristic is within a reference power range; and determine whether a temperature characteristic is within a reference temperature range.

(56) 11. The processor-based XR device of any one of clauses 9-10, wherein the processor is further configured to, responsive to determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range: update the default power profile to provide feedback for selecting the default power profile less frequently in the future; select a new power profile from the plurality of power profiles; and apply the new power profile.

(57) 12. The processor-based XR device of any one of clauses 1-11, integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a mobile phone; a cellular phone; a phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

(58) 13. A processor-based Extended Reality (XR) device, comprising: means for applying a default power profile; means for detecting an eye close event of a user of the processor-based XR device; means for selecting a test power profile from a plurality of power profiles, responsive to detecting the eye close event; means for applying the test power profile; means for measuring an operating characteristic of the processor-based XR device; means for comparing the operating characteristic with a corresponding reference range; and means for updating one or more power profiles of the plurality of power profiles based on the comparing.

(59) 14. The processor-based XR device of clause 13, further comprising: means for detecting an eye open event of the user of the processor-based XR device; and means for reverting to the default power profile, responsive to detecting the eye open event.

(60) 15. The processor-based XR device of clause 14, wherein: the means for measuring the

operating characteristic of the processor-based XR device comprises means for measuring a performance characteristic of the processor-based XR device; the means for comparing the operating characteristic with the corresponding reference range comprises means for comparing the performance characteristic with a reference performance range; and the means for updating one or more power profiles of the plurality of power profiles based on the comparing comprises: means for determining that the performance characteristic is within the reference performance range; means for measuring a power characteristic of the processor-based XR device, responsive to determining that the performance characteristic is within the reference performance range; means for comparing the power characteristic with a reference power range; means for determining that the power characteristic is within the reference power range; and means for updating the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window, responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event.

(61) 16. The processor-based XR device of clause 15, further comprising: means for determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and means for designating the test power profile as the default power profile, responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile.

(62) 17. The processor-based XR device of any one of clauses 15-16, further comprising: means for determining that the performance characteristic is not within the reference performance range; and means for reverting to the default power profile, responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event.

(63) 18. The processor-based XR device of any one of clauses 15-17, comprising: means for determining that the power characteristic is not within the reference power range; and means for reverting to the default power profile, responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event.

(64) 19. A method for providing runtime power profile tuning based on eye state, comprising: applying, by a power profile tuning circuit of a processor-based Extended Reality (XR) device, a default power profile; detecting, using an eye blink detection circuit of the processor-based XR device, an eye close event of a user of the processor-based XR device; and responsive to detecting the eye close event: selecting a test power profile from a plurality of power profiles; applying the test power profile; measuring an operating characteristic of the processor-based XR device; comparing the operating characteristic with a corresponding reference range; and updating one or more power profiles of the plurality of power profiles based on the comparing.

(65) 20. The method of clause 19, further comprising selecting the default power profile based on one or more of a use case of the processor-based XR device and a whitelist provided by an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM).

(66) 21. The method of any one of clauses 19-20, further comprising: detecting, using the eye blink detection circuit, an eye open event of the user of the processor-based XR device; and responsive to detecting the eye open event, reverting to the default power profile.

(67) 22. The method of clause 21, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating one or more power profiles of the plurality of power profiles based on the comparing comprises: determining that the performance characteristic is within the reference performance range; and responsive to determining that the performance characteristic is within the reference performance range: measuring a power characteristic of the processor-based XR device; comparing the power

characteristic with a reference power range; determining that the power characteristic is within the reference power range; and responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event, updating the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window.

(68) 23. The method of clause 22, further comprising: determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile, designating the test power profile as the default power profile.

(69) 24. The method of any one of clauses 21-23, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating one or more power profiles of the plurality of power profiles based on the comparing comprises: determining that the performance characteristic is not within the reference performance range; and responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event, reverting to the default power profile.

(70) 25. The method of any one of clauses 21-24, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating one or more power profiles of the plurality of power profiles based on the comparing comprises: determining that the performance characteristic is within the reference performance range; responsive to determining that the performance characteristic is within the reference performance range: measuring a power characteristic of the processor-based XR device; comparing the power characteristic with a reference power range; determining that the power characteristic is not within the reference power range; and responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event, reverting to the default power profile.

(71) 26. The method of any one of clauses 19-25, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a temperature characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the temperature characteristic with a reference temperature range; and updating one or more power profiles of the plurality of power profiles based on the comparing comprises: determining that the temperature characteristic is not within the reference temperature range; and responsive to determining that the temperature characteristic is not within the reference temperature range: updating the test power profile to provide feedback for selecting the test power profile less frequently in the future; selecting a new power profile from the plurality of power profiles; and applying the new power profile without waiting for an eye open event.

(72) 27. The method of any one of clauses 19-26, further comprising, prior to applying the default power profile: measuring an initial operating characteristic of the processor-based XR device; and determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range; wherein detecting the eye close event of the user of the processor-based XR device is responsive to determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range.

(73) 28. The method of clause 27, wherein determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range comprises: determining

that a performance characteristic is within a reference performance range; determining that a power characteristic is within a reference power range; and determining that a temperature characteristic is within a reference temperature range.

(74) 29. The method of any one of clauses 19-28, further comprising, prior to applying the default power profile: measuring an initial operating characteristic of the processor-based XR device; determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range; and responsive to determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range: updating the default power profile to provide feedback for selecting the default power profile less frequently in the future; selecting a new power profile from the plurality of power profiles; and applying the new power profile.

(75) 30. A non-transitory computer-readable medium having stored thereon computer-executable instructions that, when executed, cause a processor of a processor-based Extended Reality (XR) device to: apply a default power profile; detect an eye close event of a user of the processor-based XR device; and responsive to detecting the eye close event: select a test power profile from a plurality of power profiles; apply the test power profile; measure an operating characteristic of the processor-based XR device; compare the operating characteristic with a corresponding reference range; and update one or more power profiles of the plurality of power profiles based on the comparing.

Claims

1. A processor-based Extended Reality (XR) device, comprising: a processor; and an eye blink detection circuit communicatively coupled to the processor; the processor configured to: apply a default power profile; detect, using the eye blink detection circuit, an eye close event of a user of the processor-based XR device indicating a start of an eye blink window having a duration of no more than 400 milliseconds; and responsive to detecting the eye close event and during the eye blink window: select a test power profile from a plurality of power profiles; apply the test power profile; measure an operating characteristic of the processor-based XR device; compare the operating characteristic with a corresponding reference range; and update the test power profile based on the comparing; wherein each power profile of the plurality of power profiles specifies one or more of a system setting, an option, and an attribute affecting power usage of the processor-based XR device.

2. The processor-based XR device of claim 1, wherein the processor is further configured to select the default power profile based on one or more of a use case of the processor-based XR device and a whitelist provided by an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM).

3. The processor-based XR device of claim 1, wherein the processor is further configured to: detect, using the eye blink detection circuit, an eye open event of the user of the processor-based XR device indicating an end of the eye blink window; and responsive to detecting the eye open event, revert to the default power profile.

4. The processor-based XR device of claim 3, wherein: the processor is configured to measure the operating characteristic of the processor-based XR device by being configured to measure a performance characteristic of the processor-based XR device; the processor is configured to compare the operating characteristic with the corresponding reference range by being configured to compare the performance characteristic with a reference performance range; and the processor is configured to update the test power profile based on the comparing by being configured to: determine whether the performance characteristic is within the reference performance range; and responsive to determining that the performance characteristic is within the reference performance range: measure a power characteristic of the processor-based XR device; compare the power

characteristic with a reference power range; determine whether the power characteristic is within the reference power range; and responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event, update the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window.

5. The processor-based XR device of claim 4, wherein the processor is further configured to: determine whether the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile, designate the test power profile as the default power profile.

6. The processor-based XR device of claim 4, wherein the processor is further configured to, responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event, revert to the default power profile.

7. The processor-based XR device of claim 4, wherein the processor is further configured to, responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event, revert to the default power profile.

8. The processor-based XR device of claim 1, wherein: the processor is configured to measure the operating characteristic of the processor-based XR device by being configured to measure a temperature characteristic of the processor-based XR device; the processor is configured to compare the operating characteristic with the corresponding reference range by being configured to compare the temperature characteristic with a reference temperature range; and the processor is configured to update the test power profile based on the comparing by being configured to: determine whether the temperature characteristic is within the reference temperature range; and responsive to determining that the temperature characteristic is not within the reference temperature range: update the test power profile to provide feedback for selecting the test power profile less frequently in the future; select a new power profile from the plurality of power profiles; and apply the new power profile without waiting for an eye open event.

9. The processor-based XR device of claim 1, wherein the processor is further configured to, prior to applying the default power profile: measure an initial operating characteristic of the processor-based XR device; and determine whether the initial operating characteristic of the processor-based XR device is within the corresponding reference range; wherein the processor is configured to detect the eye close event of the user of the processor-based XR device responsive to determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range.

10. The processor-based XR device of claim 9, wherein the processor is configured to determine whether the initial operating characteristic of the processor-based XR device is within the corresponding reference range by being configured to: determine whether a performance characteristic is within a reference performance range; determine whether a power characteristic is within a reference power range; and determine whether a temperature characteristic is within a reference temperature range.

11. The processor-based XR device of claim 9, wherein the processor is further configured to, responsive to determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range: update the default power profile to provide feedback for selecting the default power profile less frequently in the future; select a new power profile from the plurality of power profiles; and apply the new power profile.

12. The processor-based XR device of claim 1, integrated into a device selected from the group consisting of: a set top box; an entertainment unit; a navigation device; a communications device; a fixed location data unit; a mobile location data unit; a global positioning system (GPS) device; a

mobile phone; a cellular phone; a phone; a session initiation protocol (SIP) phone; a tablet; a phablet; a server; a computer; a portable computer; a mobile computing device; a wearable computing device; a desktop computer; a personal digital assistant (PDA); a monitor; a computer monitor; a television; a tuner; a radio; a satellite radio; a music player; a digital music player; a portable music player; a digital video player; a video player; a digital video disc (DVD) player; a portable digital video player; an automobile; a vehicle component; avionics systems; a drone; and a multicopter.

13. A processor-based Extended Reality (XR) device, comprising: means for applying a default power profile; means for detecting an eye close event of a user of the processor-based XR device indicating a start of an eye blink window having a duration of no more than 400 milliseconds; means for selecting a test power profile from a plurality of power profiles, responsive to detecting the eye close event and during the eye blink window; means for applying the test power profile; means for measuring an operating characteristic of the processor-based XR device; means for comparing the operating characteristic with a corresponding reference range; and means for updating the test power profile based on the comparing; wherein each power profile of the plurality of power profiles specifies one or more of a system setting, an option, and an attribute affecting power usage of the processor-based XR device.

14. The processor-based XR device of claim 13, further comprising: means for detecting an eye open event of the user of the processor-based XR device indicating an end of the eye blink window; and means for reverting to the default power profile, responsive to detecting the eye open event.

15. The processor-based XR device of claim 14, wherein: the means for measuring the operating characteristic of the processor-based XR device comprises means for measuring a performance characteristic of the processor-based XR device; the means for comparing the operating characteristic with the corresponding reference range comprises means for comparing the performance characteristic with a reference performance range; and the means for updating the test power profile based on the comparing comprises: means for determining that the performance characteristic is within the reference performance range; means for measuring a power characteristic of the processor-based XR device, responsive to determining that the performance characteristic is within the reference performance range; means for comparing the power characteristic with a reference power range; means for determining that the power characteristic is within the reference power range; and means for updating the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window, responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event.

16. The processor-based XR device of claim 15, further comprising: means for determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and means for designating the test power profile as the default power profile, responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile.

17. The processor-based XR device of claim 15, further comprising: means for determining that the performance characteristic is not within the reference performance range; and means for reverting to the default power profile, responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event.

18. The processor-based XR device of claim 15, comprising: means for determining that the power characteristic is not within the reference power range; and means for reverting to the default power profile, responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event.

19. A method for providing runtime power profile tuning based on eye state, comprising: applying,

by a power profile tuning circuit of a processor-based Extended Reality (XR) device, a default power profile; detecting, using an eye blink detection circuit of the processor-based XR device, an eye close event of a user of the processor-based XR device indicating a start of an eye blink window having a duration of no more than 400 milliseconds; and responsive to detecting the eye close event and during the eye blink window: selecting a test power profile from a plurality of power profiles; applying the test power profile; measuring an operating characteristic of the processor-based XR device; comparing the operating characteristic with a corresponding reference range; and updating the test power profile based on the comparing; wherein each power profile of the plurality of power profiles specifies one or more of a system setting, an option, and an attribute affecting power usage of the processor-based XR device.

20. The method of claim 19, further comprising selecting the default power profile based on one or more of a use case of the processor-based XR device and a whitelist provided by an Original Equipment Manufacturer (OEM)/Original Design Manufacturer (ODM).

21. The method of claim 19, further comprising: detecting, using the eye blink detection circuit, an eye open event of the user of the processor-based XR device indicating an end of the eye blink window; and responsive to detecting the eye open event, reverting to the default power profile.

22. The method of claim 21, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating the test power profile based on the comparing comprises: determining that the performance characteristic is within the reference performance range; and responsive to determining that the performance characteristic is within the reference performance range: measuring a power characteristic of the processor-based XR device; comparing the power characteristic with a reference power range; determining that the power characteristic is within the reference power range; and responsive to determining that the power characteristic is within the reference power range and further responsive to detecting the eye open event, updating the test power profile to indicate that evaluation of the test power profile should continue during a subsequent eye blink window.

23. The method of claim 22, further comprising: determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile; and responsive to determining that the test power profile has been evaluated over multiple eye blink windows and indicates one of a performance improvement and a power improvement over the default power profile, designating the test power profile as the default power profile.

24. The method of claim 21, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating the test power profile based on the comparing comprises: determining that the performance characteristic is not within the reference performance range; and responsive to determining that the performance characteristic is not within the reference performance range and further responsive to detecting the eye open event, reverting to the default power profile.

25. The method of claim 21, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a performance characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the performance characteristic with a reference performance range; and updating the test power profile based on the comparing comprises: determining that the performance characteristic is within the reference performance range; responsive to determining that the performance characteristic is within the reference performance range: measuring a power characteristic of the processor-based XR device; comparing the power characteristic with a reference power range;

determining that the power characteristic is not within the reference power range; and responsive to determining that the power characteristic is not within the reference power range and further responsive to detecting the eye open event, reverting to the default power profile.

26. The method of claim 19, wherein: measuring the operating characteristic of the processor-based XR device comprises measuring a temperature characteristic of the processor-based XR device; comparing the operating characteristic with the corresponding reference range comprises comparing the temperature characteristic with a reference temperature range; and updating the test power profile based on the comparing comprises: determining that the temperature characteristic is not within the reference temperature range; and responsive to determining that the temperature characteristic is not within the reference temperature range: updating the test power profile to provide feedback for selecting the test power profile less frequently in the future; selecting a new power profile from the plurality of power profiles; and applying the new power profile without waiting for an eye open event.

27. The method of claim 19, further comprising, prior to applying the default power profile: measuring an initial operating characteristic of the processor-based XR device; and determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range; wherein detecting the eye close event of the user of the processor-based XR device is responsive to determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range.

28. The method of claim 27, wherein determining that the initial operating characteristic of the processor-based XR device is within the corresponding reference range comprises: determining that a performance characteristic is within a reference performance range; determining that a power characteristic is within a reference power range; and determining that a temperature characteristic is within a reference temperature range.

29. The method of claim 19, further comprising, prior to applying the default power profile: measuring an initial operating characteristic of the processor-based XR device; determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range; and responsive to determining that the initial operating characteristic of the processor-based XR device is not within the corresponding reference range: updating the default power profile to provide feedback for selecting the default power profile less frequently in the future; selecting a new power profile from the plurality of power profiles; and applying the new power profile.

30. A non-transitory computer-readable medium having stored thereon computer-executable instructions that, when executed, cause a processor of a processor-based Extended Reality (XR) device to: apply a default power profile; detect an eye close event of a user of the processor-based XR device indicating a start of an eye blink window having a duration of no more than 400 milliseconds; and responsive to detecting the eye close event and during the eye blink window: select a test power profile from a plurality of power profiles; apply the test power profile; measure an operating characteristic of the processor-based XR device; compare the operating characteristic with a corresponding reference range; and update the test power profile based on the comparing; wherein each power profile of the plurality of power profiles specifies one or more of a system setting, an option, and an attribute affecting power usage of the processor-based XR device.
